UNIT

Two

Complex numbers

- Pre-requirements for unit two : Algebraic form of a complex number.

Lesson 1 Trigonometric form of a complex number.

Lesson 2 Exponential form of a complex number (Euler form).

Lesson 3 De Moivre's theorem.

Lesson 4 Cubic roots of unity.



Pre-requirements:

Algebraic form of a complex number

The imaginary number i

The imaginary number i is defined as the number whose square equals – 1

i.e. $i^2 = -1$

* The integer powers of i gives the values i or – 1 or – i or 1 and these values are repeated frequently as the power of i increased by 4 , in general for $n \in \mathbb{Z}$, then :

$i^1 = i$	$i^2 = -1$	$i^3 = -i$	$i^4 = 1$
$i^5 = i$	$i^6 = -1$	$i^7 = -i$	$i^8 = 1$
$i^{4n+1} = i$	$i^{4n+2} = -1$	$i^{4n+3} = -i$	$i^{4n+4} = 1$

* For any positive value of a : $\sqrt{-a} = \sqrt{a \times i^2} = \sqrt{a} \times i = \sqrt{a}\, i$

For example : $\sqrt{-5} = \sqrt{5\, i^2} = \sqrt{5}\, i$, $\sqrt{-9} = \sqrt{9\, i^2} = 3\, i$ and so on.

The complex number

The complex number is the number which can be written in the form $x + y\, i$ where x , y are real numbers , $i^2 = -1$, x is called the real part and y is called the imaginary part and the form $x + y\, i$ is called the algebraic form of the complex number.

For example :

- The complex number $z_1 = 2 - 3\, i$, the real part = 2 and the imaginary part = – 3
- The complex number $z_2 = 4\, i$, the real part = zero and the imaginary part = 4 , "the number is pure imaginary".
- The complex number $z_3 = -9$, the real part = – 9 and the imaginary part = zero "the number is pure real".

Set of complex numbers C

Set of complex numbers is defined as

$C = \{x + y\, i : x \in \mathbb{R} , y \in \mathbb{R} , i^2 = -1\}$

Remarks

1. For any two complex numbers $z_1 = x_1 + y_1 i$, $z_2 = x_2 + y_2 i$, then

* $z_1 = z_2$ if and only if $x_1 = x_2$, $y_1 = y_2$

For example : If $x + y i = 5 - 4 i$, then $x = 5$, $y = -4$

* $z_1 \pm z_2 = (x_1 \pm x_2) + (y_1 \pm y_2) i$

For example : $(-1 + 4 i) - (6 - i) = -7 + 5 i$

* $z_1 \times z_2 = (x_1 x_2 - y_1 y_2) + (x_1 y_2 + x_2 y_1) i$

For example :

- $2 i (1 - 3 i) = 2 i - 6 i^2 = 6 + 2 i$
- $(3 - 5 i)(2 + i) = 6 - 7 i - 5 i^2 = 11 - 7 i$ (3 i, $-10 i$)
- $(4 - i)^2 = 16 - 8 i + i^2 = 15 - 8 i$
- $(5 - 3 i)(5 + 3 i) = 25 - 9 i^2$ (where $i^2 = -1$)
  $= 25 + 9 = 34$

2. The conjugate of the complex number $z = x + y i$ denoted as $\bar{z} = x - y i$ and notice that : $\bar{z}$ is produced by changing the sign of the imaginary part in the complex number z

For example : • If $z_1 = 3 - 4 i$, then $\overline{z_1} = 3 + 4 i$ • If $z_2 = 7 i$, then $\overline{z_2} = -7 i$

• If $z_3 = 5$, then $\overline{z_3} = 5$

3. $z + \bar{z} = (x + y i) + (x - y i) = 2 x$ (pure real)

* $z - \bar{z} = (x + i y) - (x - y i) = 2 y i$ (pure imaginary)
* $z \times \bar{z} = (x + y i)(x - y i) = x^2 + y^2$ (pure real)

For example : If $z = 3 + 2 i$, then $\bar{z} = 3 - 2 i$

and $z + \bar{z} = (3 + 2 i) + (3 - 2 i) = 6$, (pure real)

, $z - \bar{z} = (3 + 2 i) - (3 - 2 i) = 4 i$ (pure imaginary)

, $z \times \bar{z} = (3 + 2 i)(3 - 2 i) = 9 + 4 = 13$ (pure real)

4. $\dfrac{z_1}{z_2} = \dfrac{z_1 \bar{z}_2}{z_2 \bar{z}_2}$

i.e. "Multiply each of the numerator and denominator by the conjugate of z_2"

For example : $\dfrac{30 + 45 i}{1 - 2 i} = \dfrac{30 + 45 i}{1 - 2 i} \times \dfrac{1 + 2 i}{1 + 2 i} = \dfrac{30 + 105 i + 90 i^2}{1 - 4 i^2} = \dfrac{-60 + 105 i}{5} = -12 + 21 i$

5. $\overline{(z_1 \pm z_2)} = \overline{z_1} \pm \overline{z_2}$

6. $\overline{(z_1 \times z_2)} = \overline{z_1} \times \overline{z_2}$

7. $\overline{\left(\dfrac{z_1}{z_2}\right)} = \dfrac{\overline{z_1}}{\overline{z_2}}$ where $z_2 \neq 0$

8. $\bar{\bar{z}} = z$

Solving the second degree equation in one variable in C

Example

Find the solution set in C of each of the two following equations :

1. $x^2 + 4 = 0$ **2.** $x^2 + 2x + 2 = 0$

Solution

1. $x^2 + 4 = 0$

$\therefore x = \pm\sqrt{-4}$

$\therefore x^2 = -4$

$\therefore x = \pm 2i$

$\therefore$ The solution set $= \{2i, -2i\}$

2. $x^2 + 2x + 2 = 0$

$\therefore a = 1 \ , \ b = 2 \ , \ c = 2$

$$\therefore x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-2 \pm \sqrt{(2)^2 - 4(1)(2)}}{2(1)} = \frac{-2 \pm \sqrt{-4}}{2}$$

$$= \frac{-2 \pm 2i}{2} = -1 \pm i$$

$\therefore$ The solution set in $C = \{-1 + i, -1 - i\}$

Remark

If $z = x_1 + y_1 i$, is one of the roots of the equation $ax^2 + bx + c = 0$

where $a \in \mathbb{R}^*$, $b \in \mathbb{R}$, $c \in \mathbb{R}$

, then the other root is $\bar{z} = x_1 - y_1 i$

i.e. If one of the roots of the quadratic equation whose coefficients are real numbers is a complex number , then the other root is the conjugate of this complex number.

Unit 2

Lesson One

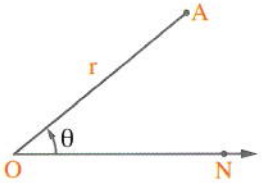
Trigonometric form of a complex number

The polar coordinates

Identifying a point (say A) in the coordinate polar system , depends on the distance of this point from a fixed point in the plane "O" and the measure of the included angle between the polar axis $\overrightarrow{ON}$ which mostly represents by a horizontal ray starts from (O) and goes right and the ray $\overrightarrow{OA}$ and the measure of this angle is positive if the direction of the angle starting from the polar axis is anti clockwise and negative if the direction is clockwise.

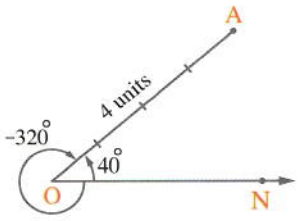
In the opposite figure :

If the distance of the point A from (O) is denoted by r and the measure of the directed $\angle$ NOA is (θ) , then the point (A) in the polar system is determined by A (r , θ)



For example :

1.

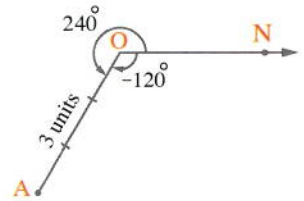


r = 4 units

$\theta = 40° \text{ or } -320°$

$\therefore$ A (4 , 40°) or (4 , – 320°)

2.



r = 3 units

$\theta = -120° \text{ or } \theta = 240°$

$\therefore$ A (3 , – 120°) or (3 , 240°)

Remark

If θ is the measure of the positive directed angle $\angle$ NOA in the polar system then , there are infinite ordered pairs represent A each in the form (r , $\theta \pm 360° n$) where $n \in z$

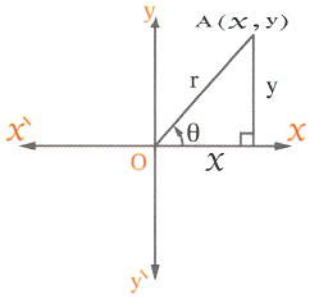
For example :

If A (6 , 30°) , then it can be represented by infinite numbers of ordered pairs as.

A (6 , – 330°) or (6 , 390°) or (6 , – 690°) or ...

Converting polar coordinates into cartesian coordinates

If the polar axis coincides with the positive direction of $\mathcal{X}-$ axis , then the polar coordinates of any point A (r , θ) can be changed into cartesian coordinate A $(\mathcal{X} , y)$ as follow : $\mathcal{X} = r \cos\theta$, $y = r \sin\theta$



i.e. $A(\mathcal{X} , y) = A(r\cos\theta , r\sin\theta)$

Remark

The cartesian coordinates of the point A $(\mathcal{X} , y)$ can be converted into the polar form A (r , θ) by evaluating r , θ as follow :

$r = \sqrt{\mathcal{X}^2 + y^2}$, $\tan\theta = \frac{y}{\mathcal{X}}$

Argand's plane

The mathematician Argand had represented the complex number $\mathcal{X} + y\,i$ graphically on the orthogonal cartesian coordinates such that the axis $\overleftrightarrow{\mathcal{X}\mathcal{X}}$ represents the real part of the complex number and the axis $\overleftrightarrow{yy}$ represents the imaginary part of the complex number. So the opposite figure on which the complex numbers are represented graphically is called Argand's plane.

In the opposite figure :

A represents the number $(4 + 2\,i)$

, B represents the number $(-2 + i)$

, C represents the number $(-4 - 3\,i)$

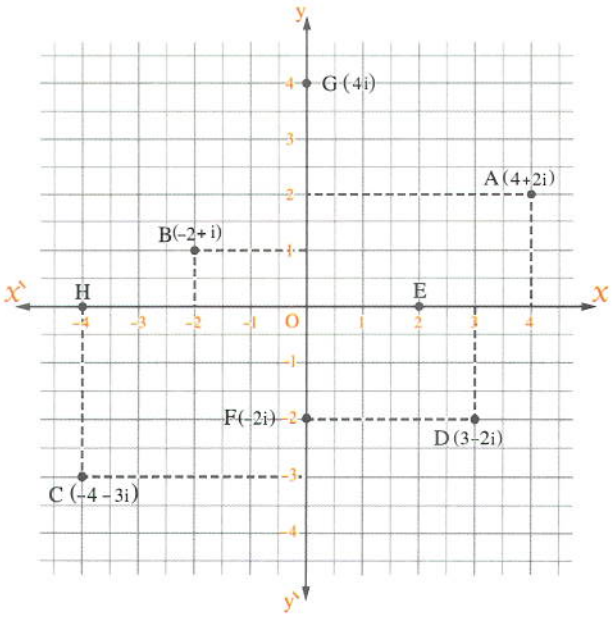
, D represents the number $(3 - 2\,i)$

, E represents the number (2)

, F represents the number $(-2\,i)$

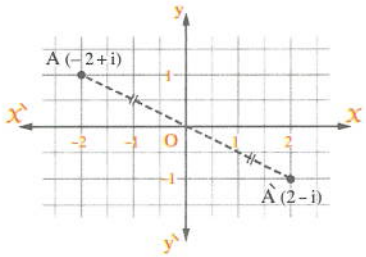
, G represents the number $(4\,i)$

and H represents the number (-4)



Remarks

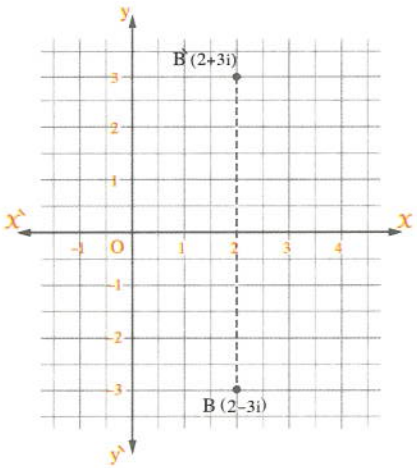
1. The two points which represent the complex number and its additive inverse on the Argand's plane are represented by two symmetric points about origin



For example :

If $z = -2 + i$ is represented by the point A and its additive inverse $-z = 2 - i$ is represented by the point $\acute{A}$, then (O) is the symmetric point of $\overline{A\acute{A}}$, (O) is the midpoint of $\overline{A\acute{A}}$

2. The two conjugate complex numbers are represented on the Argand's plane by two points symmetric points about x-axis

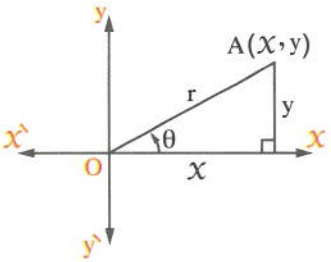


For example :

If $z = 2 - 3i$ is represented by the point B and its conjugate $\bar{z} = 2 + 3i$ is represented by the point $\acute{B}$, then $\overleftrightarrow{x'x}$ is the axis of symmetry of $\overline{B\acute{B}}$

The modulus and the amplitude and the trigonometric (Polar) form of a complex number

If a complex number $z = x + y\,i$ represents the point A (x , y) in the cartesian plane and the polar coordinates of the same point A (r , θ) , then



1. The modulus of the number z :

Is the distance between the point which represents the number z and the origin and denoted by $|z|$

i.e. $|z| = r = \sqrt{x^2 + y^2}$

2. The amplitude (argument) of the number z :

θ is called the amplitude of the number z and if $\theta \in]-\pi , \pi]$, then θ is called the principle amplitude of the number z where $\tan\theta = \frac{y}{x}$

3. The trigonometric (Polar) form of the complex number

$\because x = r\cos\theta$, $y = r\sin\theta$ $\quad\quad$ $\because z = x + y\,i$ (the algebraic form)

$\therefore z = r\cos\theta + i\,r\sin\theta$ $\quad\quad$ $\therefore z = r(\cos\theta + i\sin\theta)$ (the polar form)

Remarks

The value of θ evaluated as follow :

1. (1) If $x > 0$, $y > 0$, then θ lies in the 1st quadrant

$$\therefore \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

(2) If $x < 0$, $y > 0$, then θ lies in the 2nd quadrant

$$\therefore \theta = \pi + \tan^{-1}\left(\frac{y}{x}\right)$$

(3) If $x < 0$, $y < 0$, then θ lies in the 3rd quadrant

$$\therefore \theta = -\pi + \tan^{-1}\left(\frac{y}{x}\right)$$

(4) If $x > 0$, $y < 0$, then θ lies in the 4th quadrant

$$\therefore \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

2. (1) If $x > 0$, $y = 0$, then $\theta = 0$

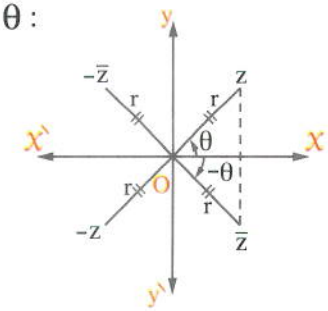
(2) If $x < 0$, $y = 0$, then $\theta = \pi$

(3) If $x = 0$, $y > 0$, then $\theta = \frac{\pi}{2}$

(4) If $x = 0$, $y < 0$, then $\theta = -\frac{\pi}{2}$

The properties of the modulus and the amplitude of a complex number

For each complex number $z = x + y\,i$ with argument (amplitude) θ :



1. $|z| \geq 0$, and $|z| = 0$ if and only if $z = 0$

2. $|z| = |\bar{z}| = |-z| = |-\bar{z}|$

i.e. The number and its conjugate and its additive inverse and the conjugate of its additive inverse are all have the same modulus.

3. $z\bar{z} = |z|^2 = |\bar{z}|^2$

4. The amplitude of the complex number can be written in infinite forms by adding complete revolutoin (2π)

i.e. The amplitude of the complex number $= \theta + 2\pi n$ where $n \in z$

5. The amplitude of the complex number doesnot change if the complex number is multiplied by positive real number.

i.e. The amplitude of z = amplitude of k z where $k \in \mathbb{R}^+$

Example 1

Find the modulus and the principle amplitude of each of the following complex numbers :

1. $z_1 = \sqrt{3} - i$ | **2.** $z_2 = \dfrac{-1}{\sqrt{2} - \sqrt{2}\,i}$

3. $z_3 = -4\,i$ | **4.** $z_4 = 6$

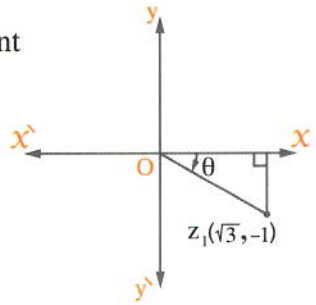
Solution

1. $\because z_1 = \sqrt{3} - i$ $\quad \therefore x = \sqrt{3}$, $y = -1$

$\therefore |z_1| = r = \sqrt{x^2 + y^2} = \sqrt{3+1} = 2$ length unit.

, $\because x > 0$, $y < 0$ $\quad \therefore \theta$ lies in the 4th quadrant

$\therefore \theta = \tan^{-1}\left(\dfrac{y}{x}\right) = \tan^{-1}\left(\dfrac{-1}{\sqrt{3}}\right) = \dfrac{-\pi}{6}$



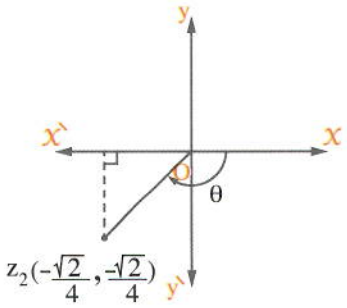
$\therefore$ The modulus of the number z_1 = 2 length unit

and the principle amplitude of the number $z_1 = \dfrac{-\pi}{6}$

2. $\because z_2 = \dfrac{-1}{\sqrt{2} - \sqrt{2}\,i} \times \dfrac{\sqrt{2} + \sqrt{2}\,i}{\sqrt{2} + \sqrt{2}\,i} = \dfrac{-\sqrt{2} - \sqrt{2}\,i}{2+2}$

$= \dfrac{-\sqrt{2} - \sqrt{2}\,i}{4} = \dfrac{-\sqrt{2}}{4} - \dfrac{\sqrt{2}}{4}\,i$

$\therefore x = \dfrac{-\sqrt{2}}{4}$, $y = \dfrac{-\sqrt{2}}{4}$



$\therefore |z_2| = r = \sqrt{x^2 + y^2} = \sqrt{\dfrac{2}{16} + \dfrac{2}{16}} = \dfrac{1}{2}$ length unit.

, $\because x < 0$, $y < 0$ $\quad \therefore \theta$ lies in the 3rd quadrant

$\therefore \theta = -\pi + \tan^{-1}\left(\dfrac{y}{x}\right) = -\pi + \tan^{-1}\left(\dfrac{-\sqrt{2}}{4} \div \dfrac{-\sqrt{2}}{4}\right) = -\pi + \tan^{-1}(1) = \dfrac{-3}{4}\pi$

$\therefore$ The modulus of the number $z_2 = \dfrac{1}{2}$ length unit.

, the principle amplitude of the number $z_2 = \dfrac{-3}{4}\pi$

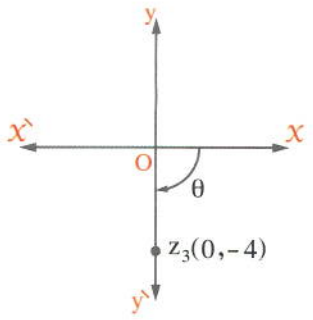
3. $\because z_3 = -4i$ $\quad \therefore x = 0 \ , \ y = -4$

$\therefore |z_3| = \sqrt{0+16} = 4$ length unit.

, $\because x = 0 \ , \ y < 0$ $\quad \therefore \theta = \frac{-\pi}{2}$

$\therefore$ The modulus of the number $z_3 = 4$

, the principle amplitude of the number $z_3 = \frac{-\pi}{2}$



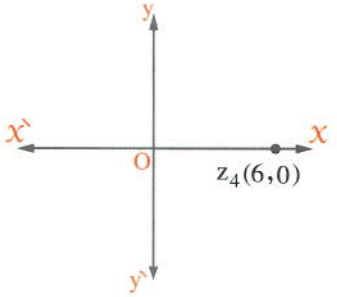
4. $\because z_4 = 6$ $\quad \therefore x = 6 \ , \ y = 0$

$\therefore |z_4| = \sqrt{36+0} = 6$ length unit.

, $\because x > 0 \ , \ y = 0$ $\quad \therefore \theta = 0$

$\therefore$ The modulus of the number $z_4 = 6$ length unit.

and the principle amplitude of the number $z_4 = 0$



Example 2

Express each of the following complex numbers in trigonometric form :

1. $z_1 = -2 + 2i$ $\quad$ **2.** $z_2 = 2\sqrt{3} - 2i$

Solution

1. $z_1 = -2 + 2i$ $\quad \therefore x = -2 \ , \ y = 2$

$\therefore |z_1| = r = \sqrt{x^2 + y^2} = \sqrt{4+4} = 2\sqrt{2}$

, $\because x < 0 \ , \ y > 0$ $\quad \therefore \theta$ lies in the 2nd quadrant

$\therefore \theta = \pi + \tan^{-1}\left(\frac{y}{x}\right) = \pi + \tan^{-1}\left(\frac{2}{-2}\right) = \pi + \tan^{-1}(-1) = \frac{3}{4}\pi$

, $\because z_1 = r(\cos\theta + i\sin\theta)$ $\quad \therefore z_1 = 2\sqrt{2}\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$

2. $z_2 = 2\sqrt{3} - 2i$ $\quad \therefore x = 2\sqrt{3} \ , \ y = -2$

$\therefore |z_2| = r = \sqrt{x^2 + y^2} = \sqrt{12+4} = 4$

$\because x > 0 \ , \ y < 0$ $\quad \therefore \theta$ lies in the 4th quadrant

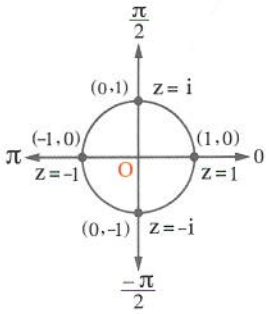
$\therefore \theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{-2}{2\sqrt{3}}\right) = \tan^{-1}\left(\frac{1}{-\sqrt{3}}\right) = \frac{-\pi}{6}$

$\therefore z_2 = r(\cos\theta + i\sin\theta) = 4\left(\cos\left(\frac{-\pi}{6}\right) + i\sin\left(\frac{-\pi}{6}\right)\right)$

Notice that :

The complex numbers 1 , i , -1 , $-i$ are represented on the Argand's plane by points lies on a circle whose centre "O" and its radius length is one unit and are produced from the intersection between this circle and the coordinate axes , the modulus of each equals one

i.e. $r = 1$ and the principle amplitude of each of them respectively : 0 , $\frac{\pi}{2}$, π , $\frac{-\pi}{2}$



and so , they can be represented in trigonometric form as follow :

$1 = \cos 0° + i \sin 0°$, $i = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2}$

, $-1 = \cos \pi + i \sin \pi$, $-i = \cos\left(\frac{-\pi}{2}\right) + i \sin\left(\frac{-\pi}{2}\right)$

Example 3

Write each of the following complex numbers in the trigonometric form :

1. 5 **2.** $3i$

3. $-\sqrt{3}$ **4.** $-\frac{1}{2} i$

Solution

1. $5 = 5 \times 1 = 5 (\cos 0° + i \sin 0°)$

2. $3i = 3 \times i = 3\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$

3. $-\sqrt{3} = \sqrt{3} \times -1 = \sqrt{3} (\cos \pi + i \sin \pi)$

4. $-\frac{1}{2} i = \frac{1}{2} \times -i = \frac{1}{2}\left(\cos\left(\frac{-\pi}{2}\right) + i \sin\left(\frac{-\pi}{2}\right)\right)$

Example 4

Express each of the following numbers in trigonometric form and also algebraic form :

1. The number z_1 whose modulus is 2 and whose principle amplitude is $\frac{\pi}{2}$

2. The number z_2 whose modulus is 3 and whose amplitude is $\frac{4\pi}{3}$

3. The number z_3 whose modulus is $2\sqrt{2}$ and whose amplitude is $315°$

4. The number z_4 whose modulus 4 , and whose amplitude $\frac{\pi}{4} + 2n\pi$ where $n \in \mathbb{Z}$

Solution

1. $\because |z_1| = 2$, $\theta = \frac{\pi}{2}$ $\quad\therefore z_1 = 2\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$ "The trigonometric form"

, $\because \cos\frac{\pi}{2} = 0$, $\sin\frac{\pi}{2} = 1$ $\quad\therefore z_1 = 2(0 + i \times 1) = 2i$ "The algebraic form"

2. $\because |z_2| = 3$, $\because$ the modulus of the number $z_2 = \frac{4\pi}{3}$

$\therefore$ The principle amplitude of the number $z_2\ (\theta) = \frac{4\pi}{3} - 2\pi = \frac{-2\pi}{3}$

$\therefore z_2 = 3\left(\cos\left(\frac{-2\pi}{3}\right) + i\sin\left(\frac{-2\pi}{3}\right)\right)$ "The trigonometric form"

, $\because \cos\left(\frac{-2\pi}{3}\right) = \cos(-120°) = \frac{-1}{2}$

, $\sin\left(\frac{-2\pi}{3}\right) = \sin(-120°) = \frac{-\sqrt{3}}{2}$

$\therefore z_2 = 3\left(\frac{-1}{2} + \frac{-\sqrt{3}}{2}i\right) = \frac{-3}{2} - \frac{3\sqrt{3}}{2}i$ "The algebraic form"

3. $\because |z_3| = 2\sqrt{2}$

, $\because$ the principle amplitude of the number $z_3\ (\theta) = 315° - 360° = -45°$

$\therefore z_3 = 2\sqrt{2}\left(\cos(-45°) + i\sin(-45°)\right)$ "The trigonometric form"

, $\because \cos(-45°) = \frac{\sqrt{2}}{2}$, $\sin(-45°) = \frac{-\sqrt{2}}{2}$

$\therefore z_3 = 2\sqrt{2}\left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i\right) = 2 - 2i$ "The algebraic form"

4. $\because |z_4| = 4$, $\because$ the amplitude of the number $z_4 = \frac{\pi}{4} + 2n\pi$, where $n \in \mathbb{Z}$

$\therefore$ The principle amplitude of the number $z_4\ (\theta) = \frac{\pi}{4}$

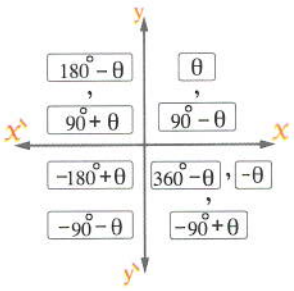
$\therefore z_4 = 4\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$ "The trigonometric form"

$\because \cos\frac{\pi}{4} = \cos 45° = \frac{\sqrt{2}}{2}$, $\sin\frac{\pi}{4} = \sin 45° = \frac{\sqrt{2}}{2}$

$\therefore z_4 = 4\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i\right) = 2\sqrt{2} + 2\sqrt{2}i$ "The algebraic form"

Convert the non standard trigonometric form into standard form [z = r (cos θ + i sin θ)]

- Determine the quadrant according to the sign in front of the two parts of the trigonometric functions , the real and the imaginary.



- In case the cosine function in the real part and the sine function in the imaginary part (the trigonometric functions in their right positions) relate the angle to 180°or 360°
- In case the sine function in the real part and the cosine function in the imaginary part (the trigonometric functions are reversed) relate the angle to 90° , − 90° , hence use the following figure.

Second quadrant

* If $z = r(-\cos\theta + i\sin\theta)$
convert it to the form
$z = r\left(\cos(180° - \theta) + i\sin(180° - \theta)\right)$

The trigonometric functions in their right positions.

* If $z = r(-\sin\theta + i\cos\theta)$
convert it to the form
$z = r\left(\cos(90° + \theta) + i\sin(90° + \theta)\right)$

The trigonometric functions are reversed.

First quadrant

* If $z = r(\cos\theta + i\sin\theta)$
still as it is :
$z = r(\cos\theta + i\sin\theta)$

The trigonometric functions in their right positions.

* If $z = r(\sin\theta + i\cos\theta)$
convert it to the form
$z = r\left(\cos(90° - \theta) + i\sin(90° - \theta)\right)$

The trigonometric functions are reversed.

Third quadrant

* If $z = r(-\cos\theta - i\sin\theta)$
convert it to the form
$z = r\left(\cos(-180° + \theta) + i\sin(-180° + \theta)\right)$

The trigonometric functions in their right positions.

* If $z = r(-\sin\theta - i\cos\theta)$
convert it to the form
$z = r\left(\cos(-90° - \theta) + i\sin(-90° - \theta)\right)$

The trigonometric functions are reversed.

Fourth quadrant

* If
$z = r(\cos\theta - i\sin\theta)$
convert it to the form
$z = r\left(\cos(-\theta) + i\sin(-\theta)\right)$

The trigonometric functions in their right positions.

* If $z = r(\sin\theta - i\cos\theta)$
convert it to the form
$z = \left(\cos(-90° + \theta) + i\sin(-90° + \theta)\right)$

The trigonometric functions are reversed.

Note that :

* We use the previous way for $r > 0$, $\theta \in [0 , 2\pi[$
* If the amplitude that we obtain $\in]-\pi , \pi]$, then it is a principle amplitude.
* If the amplitude that we obtain is not principle , then we add 360° to it or subtract 360° from it , to obtain the principle amplitude.

Example 5

Find the modulus and the principle amplitude of each of the following numbers and write the number in trigonometric form :

1. $z_1 = -4 (\cos 45° + i \sin 45°)$

2. $z_2 = 6 \left(\sin \frac{4\pi}{3} + i \cos \frac{4\pi}{3}\right)$

3. $z_3 = 8 (\cos 240° - i \sin 240°)$

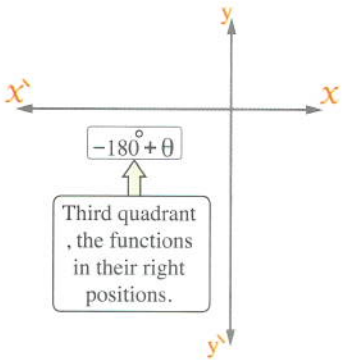
Solution

1. $z_1 = 4 (-\cos 45° - i \sin 45°)$

$\because$ The signs of trigonometric functions $(- , -)$

$\therefore$ We choose the 3rd quadrant

, $\because$ the functions in their right positions



$\therefore$ The principle amplitude of the number

$z_1 = -180° + \theta = -180° + 45° = -135°$

, the modulus of the number $z_1 = 4$

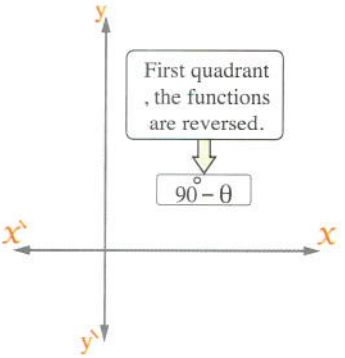
and the trigonometric form is $z_1 = 4 \left(\cos (-135°) + i \sin (-135°)\right)$

2. $z_2 = 6 \left(\sin \frac{4\pi}{3} + i \cos \frac{4\pi}{3}\right)$

, $\because$ the signs of trigonometric functions $(+ , +)$

$\therefore$ We choose the 1st quadrant

, $\because$ the functions are reversed



$\therefore$ The principle amplitude of the number z_2

$= 90° - \theta = \frac{\pi}{2} - \frac{4}{3}\pi = \frac{-5}{6}\pi$

and the modulus of the number $z_2 = 6$

and the trigonometric form $= 6 \left(\cos \left(\frac{-5\pi}{6}\right) + i \sin \left(\frac{-5\pi}{6}\right)\right)$

3. $z_3 = 8\ (\cos 240° - i \sin 240°)$

, $\because$ the signs of trigonometric functions $(+ , -)$

$\therefore$ We choose the 4th quadrant

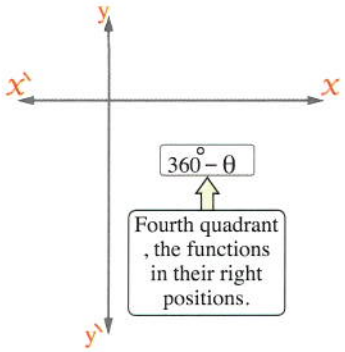
, $\because$ the functions in their right positions

$\therefore$ The principle amplitude of the number z_4

$= 360° - \theta = 360° - 240° = 120°$

, the modulus of the number $z_3 = 8$

, and the trigonometric form $z_3 = 8\ (\cos 120° + i \sin 120°)$



Example 6

If $z = r\ (\cos \theta + i \sin \theta)$ write the trigonometric form of each of the following numbers where $\theta \in]0 , \frac{\pi}{2}[$

1. $-z$ **2.** $\bar{z}$ **3.** $-\bar{z}$ **4.** $\frac{1}{z}$

Solution

1. $-z = -r\ (\cos \theta + i \sin \theta) = r\ (-\cos \theta - i \sin \theta)$

$\because x < 0$, $y < 0$

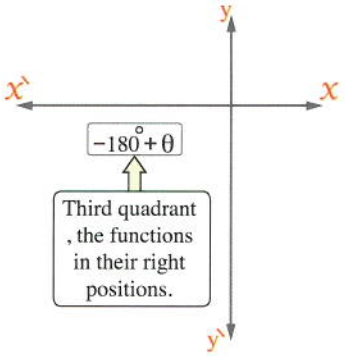
$\therefore -z$ lies in the 3rd quadrant

, $\because$ the functions in their right positions

$\therefore$ The amplitude of the number $(-z) = -180° + \theta$

and the modulus of the number $(-z) = r$

$\therefore -z = r\left(\cos (\theta - 180°) + i \sin (\theta - 180°)\right)$



2. $\bar{z} = r\ (\cos \theta - i \sin \theta)$

, $\because x > 0$, $y < 0$

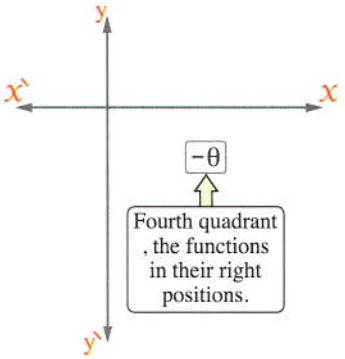
$\therefore \bar{z}$ lies in the 4th quadrant

, $\because$ the functions in their right positions

$\therefore$ The amplitude of the number $\bar{z} = -\theta$

and the modulus of the number $\bar{z} = r$

$\therefore \bar{z} = r\left(\cos (-\theta) + i \sin (-\theta)\right)$



3. $-\bar{z} = -r(\cos\theta - i\sin\theta) = r(-\cos\theta + i\sin\theta)$

$\because x < 0 \ , \ y > 0$

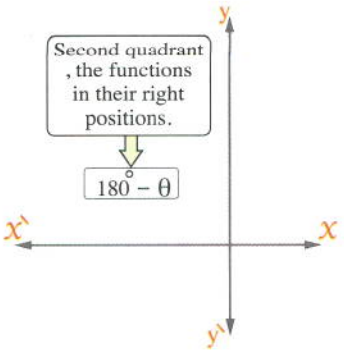
$\therefore -\bar{z}$ lies in the 2nd quadrant

, $\because$ the functions in their right positions

$\therefore$ The amplitude of the number $-\bar{z} = (180° - \theta)$

, the modulus of the number $(-\bar{z}) = r$

$\therefore -\bar{z} = r\left(\cos(180° - \theta) + i\sin(180° - \theta)\right)$



4. $\frac{1}{z} = \frac{1}{r(\cos\theta + i\sin\theta)}$ Multiply $(\cos\theta - i\sin\theta)$ by the numerator and denominator

$\therefore \frac{1}{z} = \frac{1}{r} \times \frac{1}{\cos\theta + i\sin\theta} \times \frac{(\cos\theta - i\sin\theta)}{(\cos\theta - i\sin\theta)}$

$= \frac{1}{r} \times \frac{(\cos\theta - i\sin\theta)}{(\cos^2\theta + \sin^2\theta)} = \frac{1}{r}(\cos\theta - i\sin\theta)$

Remember

$\sin^2\theta + \cos^2\theta = 1$

$\because x > 0 \ , \ y < 0$

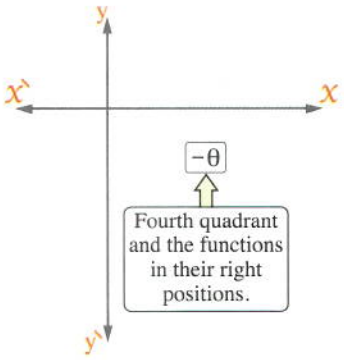
$\therefore$ The number $\frac{1}{z}$ lies in the 4th quadrant

, $\because$ the functions in their right positions

$\therefore$ The amplitude of the number $\frac{1}{z} = -\theta$

and the modulus of the number $\frac{1}{z} = \frac{1}{r}$

$\therefore \frac{1}{z} = \frac{1}{r}\left(\cos(-\theta) + i\sin(-\theta)\right)$



5. $\frac{1}{\bar{z}} = \frac{1}{r(\cos\theta - i\sin\theta)} = \frac{1}{r} \times \frac{1}{\cos\theta - i\sin\theta} \times \frac{\cos\theta + i\sin\theta}{\cos\theta + i\sin\theta}$

$= \frac{1}{r} \times \frac{\cos\theta + i\sin\theta}{\cos^2\theta + \sin^2\theta} = \frac{1}{r}(\cos\theta + i\sin\theta)$

$\therefore$ The amplitude of the number $\frac{1}{\bar{z}} = \theta$ and the modulus of the number $\frac{1}{\bar{z}} = \frac{1}{r}$

$\therefore \frac{1}{\bar{z}} = \frac{1}{r}(\cos\theta + i\sin\theta)$

Multiplying and dividing the complex numbers using the trigonometric form

If $z_1 = r_1 (\cos \theta_1 + i \sin \theta_1)$, $z_2 = r_2 (\cos \theta_2 + i \sin \theta_2)$, then :

1 $\boldsymbol{z_1 z_2 = r_1 r_2 \left(\cos (\theta_1 + \theta_2) + i \sin (\theta_1 + \theta_2)\right)}$

Proof :

$$\begin{aligned} z_1 z_2 &= r_1 (\cos \theta_1 + i \sin \theta_1) \times r_2 (\cos \theta_2 + i \sin \theta_2) \\ &= r_1 r_2 (\cos \theta_1 + i \sin \theta_1)(\cos \theta_2 + i \sin \theta_2) \\ &= r_1 r_2 (\cos \theta_1 \cos \theta_2 + i \cos \theta_1 \sin \theta_2 + i \sin \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) \\ &= r_1 r_2 (\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + i (\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2) \\ &= r_1 r_2 \left(\cos (\theta_1 + \theta_2) + i \sin (\theta_1 + \theta_2)\right) \end{aligned}$$

Notice that :

The modulus of the product of two complex numbers = the product of their moduli.

i.e. $|z_1 z_2| = r_1 r_2 = |z_1||z_2|$

and the amplitude of the product of two complex numbers = the sum of their amplitudes

i.e. The amplitude of $(z_1 z_2)$ = the amplitude of z_1 + amplitude of $z_2 = \theta_1 + \theta_2$

2 $\boldsymbol{\frac{z_1}{z_2} = \frac{r_1}{r_2} \left(\cos (\theta_1 - \theta_2) + i \sin (\theta_1 - \theta_2)\right)}$

Proof :

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{r_1 (\cos \theta_1 + i \sin \theta_1)}{r_2 (\cos \theta_2 + i \sin \theta_2)} \\ &= \frac{r_1}{r_2} \times \frac{\cos \theta_1 + i \sin \theta_1}{\cos \theta_2 + i \sin \theta_2} \times \frac{\cos \theta_2 - i \sin \theta_2}{\cos \theta_2 - i \sin \theta_2} \\ &= \frac{r_1}{r_2} \times \frac{\cos \theta_1 \cos \theta_2 - i \cos \theta_1 \sin \theta_2 + i \sin \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2}{\cos^2 \theta_2 + \sin^2 \theta_2} \\ &= \frac{r_1}{r_2} \times \frac{(\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2) + i (\sin \theta_1 \cos \theta_2 - \cos \theta_1 \sin \theta_2)}{1} \\ &= \frac{r_1}{r_2} \left(\cos (\theta_1 - \theta_2) + i \sin (\theta_1 - \theta_2)\right) \end{aligned}$$

Notice that :

The modulus of the quotient of two complex numbers = the quotient of their moduli.

i.e. $\left|\frac{z_1}{z_2}\right| = \frac{r_1}{r_2} = \frac{|z_1|}{|z_2|}$

and the amplitude of the quotient of two complex numbers = the difference of their amplitudes.

i.e. The amplitude of $\left(\frac{z_1}{z_2}\right)$ = The amplitude of z_1 – the amplitude of $z_2 = \theta_1 - \theta_2$

Important remark

To use the multiplication and division rules in the complex numbers using the trigonometric forms they must be in the standard trigonometric form.

i.e. In the form $r(\cos\theta + i\sin\theta)$

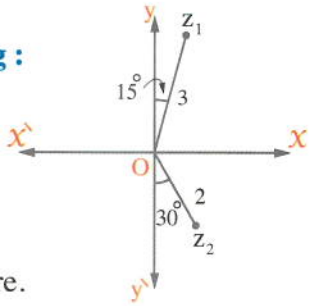
Example 7

Find $z_1 z_2$, $\frac{z_1}{z_2}$ in the trigonometric form in each of the following :

1. $z_1 = 2(\cos 60° + i \sin 60°)$, $z_2 = 4(\cos 150° + i \sin 150°)$

2. $z_1 = 2\left(\cos\frac{\pi}{5} + i\sin\frac{\pi}{5}\right)$, $z_2 = 8\left(\cos\frac{-2\pi}{5} + i\sin\frac{-2\pi}{5}\right)$

3. z_1 , z_2 are represented on Argand's diagram as the opposite figure.



Solution

1. $z_1 z_2 = 2(\cos 60° + i \sin 60°) \times 4(\cos 150° + i \sin 150°)$

$= 8\left(\cos(60° + 150°) + i\sin(60° + 150°)\right)$

$= 8(\cos 210° + i \sin 210°) = 8\left(\cos(-150°) + i\sin(-150°)\right)$

, $\frac{z_1}{z_2} = \frac{2(\cos 60° + i \sin 60°)}{4(\cos 150° + i \sin 150°)}$

$= \frac{1}{2}\left(\cos(60° - 150°) + i\sin(60° - 150°)\right)$

$= \frac{1}{2}\left(\cos(-90°) + i\sin(-90°)\right)$

2. $z_1 z_2 = 2\left(\cos\frac{\pi}{5} + i\sin\frac{\pi}{5}\right) \times 8\left(\cos\left(\frac{-2\pi}{5}\right) + i\sin\left(\frac{-2\pi}{5}\right)\right)$

$= 2 \times 8\left(\cos\left(\frac{\pi}{5} + \frac{-2\pi}{5}\right)\right) + i\sin\left(\frac{\pi}{5} + \left(\frac{-2\pi}{5}\right)\right) = 16\left(\cos\left(\frac{-\pi}{5}\right) + i\sin\left(\frac{-\pi}{5}\right)\right)$

$$, \frac{z_1}{z_2} = \frac{2\left(\cos\left(\frac{\pi}{5}\right) + i \sin\left(\frac{\pi}{5}\right)\right)}{8\left(\cos\left(\frac{-2\pi}{5}\right) + i \sin\left(\frac{-2\pi}{5}\right)\right)}$$

$$= \frac{1}{4}\left(\cos\left(\frac{\pi}{5} - \left(\frac{-2\pi}{5}\right)\right) + i \sin\left(\frac{\pi}{5} - \left(\frac{-2\pi}{5}\right)\right)\right)$$

$$= \frac{1}{4}\left(\cos\left(\frac{3\pi}{5}\right) + i \sin\left(\frac{3\pi}{5}\right)\right)$$

3. From the figure :

$z_1 = 3\ (\cos 75° + i \sin 75°)$

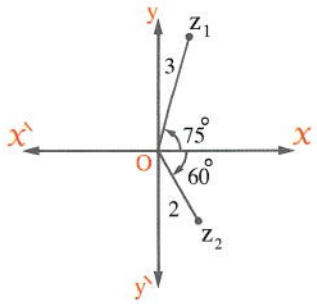
$, z_2 = 2\left(\cos(-60°) + i \sin(-60°)\right)$

$\therefore z_1 z_2 = 3 \times 2\left(\cos(75° - 60°) + i \sin(75° - 60°)\right)$

$= 6\left(\cos 15° + i \sin 15°\right)$

$$, \frac{z_1}{z_2} = \frac{3}{2}\left(\cos(75° + 60°) + i \sin(75° + 60°)\right)$$

$$= \frac{3}{2}(\cos 135° + i \sin 135°)$$



Results

1. If $z = r(\cos\theta + i \sin\theta)$, then :

(1) $\frac{1}{z} = \frac{1}{r}\left(\cos(-\theta) + i \sin(-\theta)\right)$ (2) $z^2 = r^2\left(\cos(2\theta) + i \sin(2\theta)\right)$

2. If $z_1 = r_1(\cos\theta_1 + i \sin\theta_1)$, $z_2 = r_2(\cos\theta_2 + i \sin\theta_2)$, ... ,

$z_n = r_n(\cos\theta_n + i \sin\theta_n)$

, then $z_1 z_2 \ldots\ldots z_n = r_1 r_2 \ldots\ldots r_n\left(\cos(\theta_1 + \theta_2 + \ldots\ldots + \theta_n) + i \sin(\theta_1 + \theta_2 + \ldots\ldots \theta_n)\right)$

i.e. So , we can generalize the product rule on definite numbers of the complex numbers.

If $z_1 = z_2 = z_3 = \ldots\ldots z$

then $z^n = r^n\left(\cos(n\theta) + i \sin(n\theta)\right)$

from that we find $|z^n| = |z|^n$

Example 8

If $z_1 = 2(\cos 120° - i \sin 120°)$, $z_2 = 3(\sin 150° + i \cos 150°)$

find in the trigonometric and algebraic form :

1. $z_1 z_2$ **2.** $\frac{1}{z_1}$

3. z_1^2 **4.** z_2^4

Solution

At the beginning , putting each of z_1 , z_2 in the trigonometric form

$\because z_1 = 2(\cos 120° - i \sin 120°)$

, $\because$ the signs of the trigonometric functions are (+ , –)

$\therefore$ We choose the fourth quadrant.

, $\because$ the trigonometric functions are correct

$\therefore$ The principle amplitude for the number $z_1 = -120°$

$\therefore z_1 = 2\left(\cos(-120°) + i \sin(-120°)\right)$

, $\because z_2 = 3(\sin 150° + i \cos 150°)$

, $\because$ the signs of the trigonometric functions are (+ , +)

$\therefore$ We choose the first quadrant.

, $\because$ the trigonometric functions are switched

$\therefore$ The principle amplitude for the number $z_2 = 90° - 150° = -60°$

$\therefore z_2 = 3\left(\cos(-60°) + i \sin(-60°)\right)$

1. $z_1 z_2 = 2 \times 3\left(\cos(-120° - 60°) + i \sin(-120° - 60°)\right)$

$= 6\left(\cos(-180°) + i \sin(-180°)\right) = 6(\cos 180° + i \sin 180°)$"Trigonometric form"

, $z_1 z_2 = 6(-1 + i \times 0) = -6$ "Algebraic form"

2. $\frac{1}{z_1} = \frac{1}{2\left(\cos(-120°) + i \sin(-120°)\right)}$

$= \frac{1}{2}(\cos 120° + i \sin 120°)$ " Trigonometric form"

$\therefore \frac{1}{z_1} = \frac{1}{2}\left(\frac{-1}{2} + \frac{\sqrt{3}}{2} i\right) = \frac{-1}{4} + \frac{\sqrt{3}}{4} i$ "Algebraic form"

3. $z_1^2 = 2^2\left(\cos 2(-120°) + i \sin 2(-120°)\right)$

$= 4\left(\cos(-240°) + i \sin(-240°)\right)$

$= 4(\cos 120° + i \sin 120°)$ "Trigonometric form"

, $z_1^2 = 4\left(\frac{-1}{2} + \frac{\sqrt{3}}{2} i\right) = -2 + 2\sqrt{3}\, i$ "Algebraic form"

4. $z_2^4 = 3^4\left(\cos 4(-60°) + i \sin 4(-60°)\right)$

$= 81\left(\cos(-240°) + i \sin(-240°)\right)$

$= 81(\cos 120° + i \sin 120°)$ "Trigonometric form"

$= 81\left(\frac{-1}{2} + \frac{\sqrt{3}}{2} i\right) = \frac{-81}{2} + \frac{81\sqrt{3}}{2} i$ "Algebraic form"

Example 9

If z is a complex number

The principle amplitude for the number $(-z)$ equals $\frac{-5\pi}{6}$

Find the principle amplitude for the number $\left(\frac{\bar{z}}{i}\right)$

Solution

Let the principle amplitude for the complex number z is θ

$\therefore$ The principle amplitude for the number $(-z)$ is $\theta - \pi$

$\therefore \theta - \pi = \frac{-5\pi}{6}$ $\quad\quad$ $\therefore \theta = \frac{\pi}{6}$

, the principle amplitude for the number $(\bar{z}) = \frac{-\pi}{6}$

$\therefore$ The principle amplitude for the number $\left(\frac{\bar{z}}{i}\right)$ = (amplitude of $\bar{z}$) – (amplitude of i)

$= -\frac{\pi}{6} - \frac{\pi}{2} = \frac{-2\pi}{3}$

Example 10

If $z = 2(\cos\theta + i \sin\theta)$ where $\theta \in \left[\pi, \frac{3}{2}\pi\right[$, $\tan\theta = \frac{3}{4}$

Find the trigonometric form and algebraic form for each of : $\frac{1}{z}$, z^{-2} , $\frac{1}{\bar{z}}$

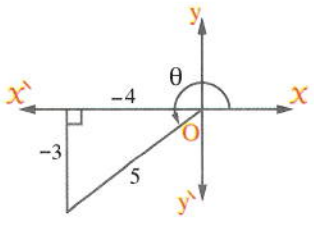
Solution

$\because z = 2(\cos\theta + i \sin\theta)$

$\therefore \frac{1}{z} = \frac{1}{2}\left(\cos(-\theta) + i \sin(-\theta)\right)$ " Trigonometric form"

$= \frac{1}{2}(\cos\theta - i \sin\theta) = \frac{1}{2}\left(-\frac{4}{5} + \frac{3}{5} i\right)$

$\therefore \frac{1}{z} = -\frac{2}{5} + \frac{3}{10} i$ "Algebraic form"



$\because z^2 = 4(\cos\theta + i\sin\theta)^2$

$\therefore z^2 = 4(\cos 2\theta + i\sin 2\theta)$

$\therefore \frac{1}{z^2} = \frac{1}{4}[\cos(-2\theta) + i\sin(-2\theta)]$ " Trigonometric form"

$= \frac{1}{4}[\cos 2\theta - i\sin 2\theta] = \frac{1}{4}[2\cos^2\theta - 1 - i \times 2\sin\theta\cos\theta]$

$= \frac{1}{4}\left(\frac{32}{25} - 1 - i \times 2 \times \frac{-3}{5} \times \frac{-4}{5}\right) = \frac{1}{4}\left(\frac{7}{25} - \frac{24}{25}i\right)$

$\therefore z^{-2} = \frac{7}{100} - \frac{6}{25}i$ "Algebraic form"

, $\because \bar{z} = 2(\cos\theta - i\sin\theta) = 2\left(\cos(-\theta) + i\sin(-\theta)\right)$

$\therefore \frac{1}{\bar{z}} = \frac{1}{2}\left(\cos -(-\theta) + i\sin -(-\theta)\right)$

$= \frac{1}{2}(\cos\theta + i\sin\theta)$ "Trigonometric form"

$= \frac{1}{2}\left(\frac{-4}{5} - \frac{3}{5}i\right)$

$\therefore \frac{1}{\bar{z}} = \frac{-2}{5} - \frac{3}{10}i$ "Algebraic form"

Remarks

1. If $z = \frac{x + iy}{x - iy}$, then $|z| = 1$ (where the numerator is conjugate of denominator)

For example : $\left|\frac{3+i}{3-i}\right| = 1$, $\left|\frac{\sqrt{a+2} - 4i}{\sqrt{a+2} + 4i}\right| = 1$

2. If z is a complex number , its amplitude is θ , then the number (i z) is the image of the number (z) by rotation 90° about the origin so $|iz| = |z|$ and amplitude (i z) = 90° + θ

Unit 2

Lesson Two

Exponential form of a complex number (Euler form)

Prelude

Before studying the exponential form of the complex number (Eular form) , you should know some series known as (Maclaurin series) or (Maclaurin expansion) that are used to express a function as a power series of the variable including in the function , next part shows some Maclaurin expansion of some functions that will be used in studying the complex numbers written in the exponential form.

* The sine function : $y = \sin x$: $\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n}{\lfloor 2n+1} x^{2n+1}$

i.e. $\sin x = \frac{x}{\lfloor 1} - \frac{x^3}{\lfloor 3} + \frac{x^5}{\lfloor 5} - \ldots\ldots\ldots$

* The cosine function : $y = \cos x$: $\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{\lfloor 2n} x^{2n}$

i.e. $\cos x = 1 - \frac{x^2}{\lfloor 2} + \frac{x^4}{\lfloor 4} - \ldots\ldots\ldots$

* The exponential function : $y = e^x$: $e^x = \sum_{n=0}^{\infty} \frac{x^n}{\lfloor n}$

i.e. $e^x = 1 + \frac{x}{\lfloor 1} + \frac{x^2}{\lfloor 2} + \frac{x^3}{\lfloor 3} + \ldots\ldots\ldots$

Exponential form of a complex number (Euler form)

Notice that :

$$e^{ix} = 1 + \frac{ix}{\lfloor 1} + \frac{i^2x^2}{\lfloor 2} + \frac{i^3x^3}{\lfloor 3} + \frac{i^4x^4}{\lfloor 4} + \ldots\ldots\ldots$$

$$= 1 + \frac{ix}{\lfloor 1} - \frac{x^2}{\lfloor 2} - \frac{ix^3}{\lfloor 3} + \frac{x^4}{\lfloor 4} - \ldots\ldots\ldots$$

$$= \left(1 - \frac{x^2}{\lfloor 2} + \frac{x^4}{\lfloor 4} - \ldots\ldots\ldots\right) + i\left(\frac{x}{\lfloor 1} - \frac{x^3}{\lfloor 3} + \ldots\ldots\ldots\right)$$

$$e^{ix} = \cos x + i \sin x$$

i.e. The complex number $z = x + i\,y = r(\cos\theta + i \sin\theta)$ can be written

In the form : $z = r\,e^{\theta i}$

it is called Euler form of the complex number where θ is in radian measure.

Example 1

Put each of the following complex numbers in the exponential form (Euler's form) :

1. $z_1 = 1 - \sqrt{3}\,i$ **2.** $z_2 = \frac{1}{1+i}$

Solution

1. $z_1 = 1 - \sqrt{3}\,i$ $\quad\therefore x = 1 \quad , \quad y = -\sqrt{3}$

$\because r = |z_1| = \sqrt{x^2 + y^2}$ $\quad\therefore r = \sqrt{1+3} = 2$

, $\because x > 0 \quad , \quad y < 0$ $\quad\therefore z_1$ lies in the 4th quadrant

$\therefore \theta = \tan^{-1}\frac{y}{x} = \tan^{-1}\left(\frac{-\sqrt{3}}{1}\right)$ $\quad\therefore \theta = \frac{-\pi}{3}$

$\therefore z_1 = 2\,e^{\frac{-\pi}{3}i}$ (Exponential form)

2. $z_2 = \frac{1}{1+i} \times \frac{1-i}{1-i} = \frac{1-i}{1+1} = \frac{1}{2} - \frac{1}{2}\,i$ $\quad\therefore x = \frac{1}{2}\ , y = \frac{-1}{2}$

, $\because r = |z_2| = \sqrt{x^2 + y^2}$ $\quad\therefore r = \sqrt{\frac{1}{4} + \frac{1}{4}} = \frac{\sqrt{2}}{2}$

, $\because x > 0\ , y < 0$ $\quad\therefore z_2$ lies in the 4th quadrant

$\therefore \theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{-1}{2} \div \frac{1}{2}\right) = \tan^{-1}(-1)$

$\therefore \theta = \frac{-\pi}{4}$

$\therefore z_2 = \frac{\sqrt{2}}{2}\,e^{\frac{-\pi}{4}i}$ (Exponential form)

Another solution :

$\because z_2 = \frac{1}{1+i}$, let $1 + i = z$ $\quad\therefore |z| = \sqrt{1+1} = \sqrt{2}$

, $\because x > 0\ , y > 0$ $\quad\therefore z$ lies in the 1st quadrant

$\therefore \theta = \tan^{-1}(1) = \frac{\pi}{4}$ $\quad\therefore z = \sqrt{2}\,e^{\frac{\pi}{4}i}$

$\therefore z_2 = \frac{1}{\sqrt{2}\,e^{\frac{\pi}{4}i}} = \frac{\sqrt{2}}{2}\,e^{\frac{-\pi}{4}i}$

Remember that

$\frac{1}{a^n} = a^{-n}$, where $a \neq 0$

Example 2

Put the complex number $z = \frac{2(1+i)}{1-i}$ in the trigonometric form then find the number z in the exponential form.

Solution

$$z = \frac{2(1+i)}{1-i} \times \frac{1+i}{1+i} = \frac{2(1+i)^2}{1+1} = \frac{2(1+2i+i^2)}{2} = 2i$$

$\therefore z = 2 \times i = 2\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$ (Trigonometric form)

$\therefore r = |z| = 2 \ , \theta = \frac{\pi}{2}$ $\qquad$ $\therefore z = 2\,e^{\frac{\pi}{2}i}$ (Exponential form)

Remarks

$i = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} = e^{\frac{\pi}{2}i}$, $-i = \cos\frac{-\pi}{2} + i\sin\frac{-\pi}{2} = e^{\frac{-\pi}{2}i}$

, $1 = \cos 0 + i\sin 0 = e^{0\,i}$, $-1 = \cos\pi + i\sin\pi = e^{\pi i}$

Example 3

Put the complex number $z = 4\left(\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}\right)$ in the exponential form [Euler's form].

Solution

$\because z = 4\left(\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}\right)$

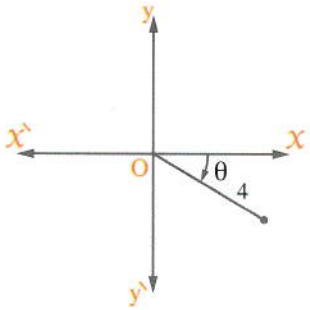
$\therefore |z| = r = 4$, $\because x > 0$, $y < 0$

$\therefore$ The number z lies in the 4th quadrant

$\therefore \theta = -\frac{\pi}{3}$

$\therefore z = 4\left(\cos\left(\frac{-\pi}{3}\right) + i\sin\left(\frac{-\pi}{3}\right)\right)$ (Trigonometric form)

$\therefore z = r\,e^{\theta i} = 4\,e^{\frac{-\pi}{3}i}$ (Exponential form)



Example 4

Find the algebraic form for each of the following :

1. $z = \sqrt{3}\,e^{\frac{5\pi}{3}i}$ $\qquad$ **2.** $z = e^{3+\frac{\pi}{6}i}$

Solution

1. $z = \sqrt{3}\, e^{\frac{5\pi}{3} i}$ $\quad \therefore r = |z| = \sqrt{3}$, $\theta = \frac{5\pi}{3} - 2\pi = -\frac{\pi}{3}$

$\therefore z = r(\cos\theta + i \sin\theta) = \sqrt{3}\left(\cos\left(\frac{-\pi}{3}\right) + i \sin\left(\frac{-\pi}{3}\right)\right)$ (Trigonometric form)

$\therefore z = \sqrt{3}\left(\frac{1}{2} + i \times -\frac{\sqrt{3}}{2}\right) = \frac{\sqrt{3}}{2} - \frac{3}{2} i$ (Algebraic form)

2. $z = e^{3 + \frac{\pi}{6} i} = e^3 \times e^{\frac{\pi}{6} i}$ $\quad \therefore r = |z| = e^3$, $\theta = \frac{\pi}{6}$

$\therefore z = r(\cos\theta + i \sin\theta) = e^3\left(\cos\frac{\pi}{6} + i \sin\frac{\pi}{6}\right)$ (Trigonometric form)

$\therefore z = e^3\left(\frac{\sqrt{3}}{2} + i \times \frac{1}{2}\right) = \frac{\sqrt{3}}{2} e^3 + \frac{1}{2} e^3 i$ (Algebraic form)

Example 5

Prove that :

1. $\cos\theta = \frac{e^{\theta i} + e^{-\theta i}}{2}$ $\qquad$ **2.** $\sin\theta = \frac{e^{\theta i} - e^{-\theta i}}{2i}$

3. $e^{3 + \pi i} - e^{-\pi i} = (1 - e)(1 + e + e^2)$

Solution

$\because e^{\theta i} = \cos\theta + i \sin\theta$ $\qquad$ (1)

$\therefore e^{-\theta i} = \cos(-\theta) + i \sin(-\theta)$

$\therefore e^{-\theta i} = \cos\theta - i \sin\theta$ $\qquad$ (2)

1. Adding (1) , (2) :

$\therefore e^{\theta i} + e^{-\theta i} = 2\cos\theta$ $\qquad \therefore \cos\theta = \frac{e^{\theta i} + e^{-\theta i}}{2}$

2. Subtracting (2) from (1) :

$\therefore e^{\theta i} - e^{-\theta i} = 2i \sin\theta$ $\qquad \therefore \sin\theta = \frac{e^{\theta i} - e^{-\theta i}}{2i}$

3. $\because e^{3 + \pi i} = e^3 \times e^{\pi i}$

$= e^3(\cos\pi + i \sin\pi) = e^3(-1 + 0) = -e^3$

, $\because e^{-\pi i} = \cos(-\pi) + i \sin(-\pi) = -1 + 0 = -1$

$\therefore e^{3 + \pi i} - e^{-\pi i} = -e^3 + 1 = 1 - e^3 = (1 - e)(1 + e + e^2)$

Multiplying and dividing the complex numbers using the exponential form

1. If $z_1 = r_1 e^{\theta_1 i}$, $z_2 = r_2 e^{\theta_2 i}$, then :

(1) $z_1 z_2 = r_1 e^{\theta_1 i} \times r_2 e^{\theta_2 i} = r_1 r_2 e^{\theta_1 i + \theta_2 i}$ **i.e.** $z_1 z_2 = r_1 r_2 e^{(\theta_1 + \theta_2) i}$

(2) $\frac{z_1}{z_2} = \frac{r_1 e^{\theta_1 i}}{r_2 e^{\theta_2 i}} = \frac{r_1}{r_2} e^{\theta_1 i - \theta_2 i}$ **i.e.** $\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{(\theta_1 - \theta_2) i}$

2. For any number $z = r e^{\theta i}$ then $z^n = r^n e^{\theta n i}$ where $n \in \mathbb{Z}$

Example 6

If $z_1 = 10 (\cos 310° + i \sin 310°)$, $z_2 = 5 (\cos 100° + i \sin 100°)$, find in the exponential form each of :

1. $\frac{z_1}{z_2}$ **2.** $\frac{z_2}{z_1}$

Solution

$\because z_1 = 10 (\cos 310° + i \sin 310°) = 10 \left(\cos (-50°) + i \sin (-50°)\right) = 10 e^{\frac{-5}{18} \pi i}$

, $z_2 = 5 (\cos 100° + i \sin 100°) = 5 e^{\frac{5}{9} \pi i}$

1. $\frac{z_1}{z_2} = \frac{10}{5} e^{\left(\frac{-5}{18} - \frac{5}{9}\right) \pi i} = 2 e^{-\frac{5}{6} \pi i}$

2. $\frac{z_2}{z_1} = \left(\frac{z_1}{z_2}\right)^{-1} = \frac{1}{2} e^{\frac{5\pi}{6} i}$

Example 7

Put the complex number $z = \left(\sqrt{3} + i\right)^4$ in Euler's form

Solution

Put $z_1 = \sqrt{3} + i$

$\therefore |z_1| = \sqrt{3 + 1} = 2$

, $\because x > 0$, $y > 0$ $\therefore z_1$ lies in the 1st quadrant

$\therefore \theta_1 = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$

$\therefore z_1 = 2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right) = 2 e^{\frac{\pi}{6} i}$ $\therefore z = z_1^4 = 2^4 e^{\frac{4\pi}{6} i} = 16 e^{\frac{2\pi}{3} i}$